

HANDS ON EXERCISE – DESIGN PRINCIPLES AND PATTERNS

Exercise 1: Implementing the Singleton Pattern

CODE:

```
1 using System;
2
3 public sealed class Logger
4 {
5     private static Logger instance;
6
7     private Logger()
8     {
9     }
10
11     public static Logger GetInstance()
12     {
13         if (instance == null)
14         {
15             instance = new Logger();
16         }
17         return instance;
18     }
19
20     public void Log(string message)
21     {
22         Console.WriteLine($"Log: {message}");
23     }
24 }
25
26 class Program
27 {
28     static void Main()
29     {
30         Logger logger1 = Logger.GetInstance();
31         Logger logger2 = Logger.GetInstance();
32
33         logger1.Log("Application Started");
34
35         Console.WriteLine(
36             ReferenceEquals(logger1, logger2)
37                 ? "Only one instance created"
38                 : "Different instances created"
39         );
40     }
41 }
```

OUTPUT:

```
Log: Application Started
Only one instance created

...Program finished with exit code 0
Press ENTER to exit console.
```

Exercise 2: Implementing the Factory Method Pattern

CODE:

```
main.cs
1 using System;
2
3 public interface IDocument
4 {
5     void Open();
6 }
7
8 public class WordDocument : IDocument
9 {
10     public void Open()
11     {
12         Console.WriteLine("Opening Word Document");
13     }
14 }
15
16 public class PdfDocument : IDocument
17 {
18     public void Open()
19     {
20         Console.WriteLine("Opening PDF Document");
21     }
22 }
23
24 public abstract class DocumentFactory
25 {
26     public abstract IDocument CreateDocument();
27 }
28
29 public class WordFactory : DocumentFactory
30 {
31     public override IDocument CreateDocument()
32     {
33         return new WordDocument();
34     }
35 }
36
37 public class PdfFactory : DocumentFactory
38 {
39     public override IDocument CreateDocument()
40     {
41         return new PdfDocument();
42     }
43 }
44
45 class Program
46 {
47     static void Main()
48     {
49         DocumentFactory factory = new WordFactory();
50         IDocument doc = factory.CreateDocument();
51         doc.Open();
52
53         factory = new PdfFactory();
54         doc = factory.CreateDocument();
55         doc.Open();
56     }
57 }
```

OUTPUT:

```
Opening Word Document
Opening PDF Document

...Program finished with exit code 0
Press ENTER to exit console.
```